Name _____ Period _____

GEOMETRY • COORDINATE REVIEW

Midpoint, Distance & Slope

Vocabulary · fill in the missing word

Midpoint — the point halfway between the endpoints; find it by _____ the x's and the y's

Distance — the length of a segment, found with the _____ Theorem

Slope — the steepness of a line: _____ over run

Parallel / perpendicular — parallel lines have the _____ slope; perpendicular slopes multiply to _____

Where's the middle?	How long is it?	How steep is it?
$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Average the x's and the y's	The Pythagorean Theorem	Rise over run

We Do: Practice

1. P(−5, 2) and Q(3, 8). Find the midpoint, distance, and slope.

Midpoint _____ Distance _____ Slope _____

2. R(0, −3) and S(5, 9). Find the midpoint, distance, and slope.

Midpoint _____ Distance _____ Slope _____

3. T(3, −2) and U(3, 6). Find the midpoint, distance, and slope.

Midpoint _____ Distance _____ Slope _____

4. M(3, −1) is the midpoint of $\overline{AB}$. A is (7, 4). Find B.

5. Segment 1 goes through (1, 2) and (4, 8). Segment 2 goes through (0, 5) and (4, 3). Are they parallel, perpendicular, or neither? Show the slopes.

Name _____ Period _____

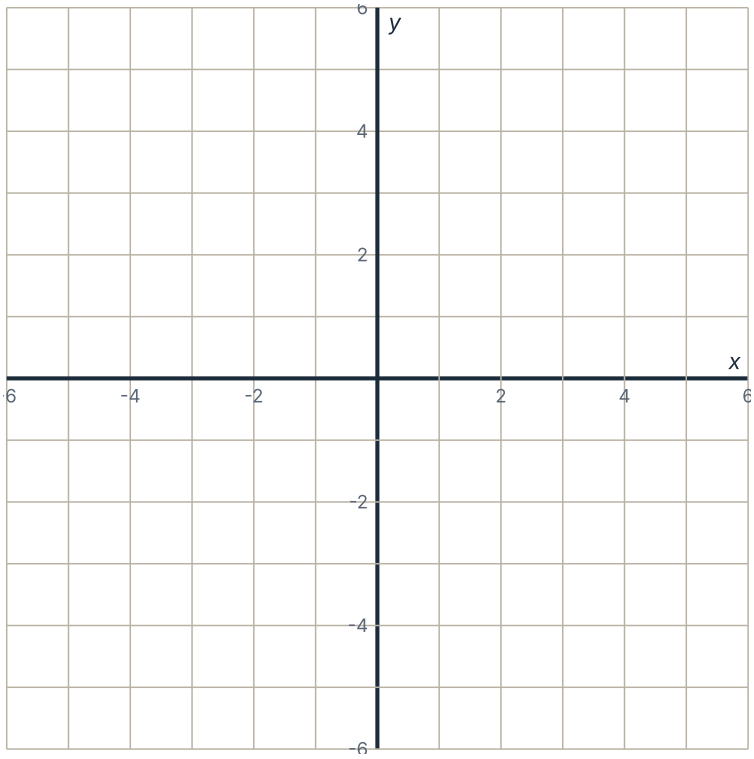
PARTNER TASK • 12 MINUTES

Is ABCD a square?

A(−3, 1) B(1, 4) C(4, 0) D(0, −3)

Partner A: _____ (finds the lengths) Partner B: _____ (finds the slopes)

1. Plot ABCD.



2–3. Sides

Side	Length (Partner A)	Slope (Partner B)
AB		
BC		
CD		
DA		

Swap and check each other's work before step 5.

4. Together: find the midpoint of each diagonal.

Midpoint of $\overline{AC}$:

Midpoint of $\overline{BD}$:

What does it tell you that the diagonals have the same midpoint?

_____

5. Is ABCD a square? Use your numbers to prove it.

Done early? Find the length of both diagonals. What do you notice?